



FIRST SEMESTER 2026-27
COURSE HANDOUT (PART-II)

Date: 31-07-2026

In addition to part I (General handout for all courses appended to the timetable) this portion gives further details regarding the course.

Course No. : CHEM F214
Course Title : Inorganic Chemistry I
Instructor-in-charge : SUROJIT PANDE

1. Course Description:

Structure of molecules: VSEPR model; ionic crystal structure, structure of complex solids; concepts of inorganic chemistry: electronegativity, acid-base chemistry, chemistry of aqueous and non aqueous solvents; descriptive chemistry of some elements: periodicity, chemistry of transition metals, halogens and noble gases; inorganic chains, rings, cages and clusters.

2. Scope and Objective of the Course:

The course is designed in consideration to growing the basic level knowledge and understanding in inorganic chemistry. It includes electronegativity, detailed study of acid-base chemistry, non-aqueous solution chemistry, concept of redox chemistry, shape and structure of inorganic covalent molecules using VSEPR model and bonding of ionic compounds in solid state. It includes the chemistry of halogen and noble gases and main group elements. There will have thorough discussion of chemistry of macromolecules containing inorganic chains, rings and cages. This course will attempt to provide the students with sufficient basic knowledge of the structure and reactivity of inorganic systems to ensure a more comprehensive understanding.

By the end of semester, you will be able to learn –

- Different scales to define electronegativity; rationalization of unusual reactivity
- Shape and structure of different covalent molecules and ionic compounds
- Measures of acid-base strength in gas and solution phase; Relation between hardness and softness, Symbiosis
- Chemistry of protonic and aprotic solvents; Redox stability; Diagrammatic presentation of potential data
- Use (or not) of d orbitals by nonmetals; Reactivity and d orbital participation
- Positively charged halogens; Pseudohalogens
- Numerous examples in nature of inorganic chains, rings and cages – their structures and chemistry

3. Text Book:

T1. Inorganic Chemistry by Huheey J. E., Keiter, E. A. Keiter, R. L. Keiter, O. K. Medhi, 4th ed., Pearson Education





4. Reference Books:

R1. Inorganic Chemistry by Shriver & Atkins, (4th edition)

R2. Advanced Inorganic Chemistry by Cotton F.A., Wilkinson G., Murillo, C.A., Bochmann, M., 6th ed., John Wiley and Sons, New York (2003).

5. Course plan:

Module no.	Broad topics	Lecture Session	Reference (Chapter & Page no. of T1)
1 (1-4)	Chemical Bonding: Ionic Bond and Solid State Structure	Ionic Bond, Close Packing; Lattice energy, Size effects, Covalent character in predominantly ionic bonds, Imperfections in crystals, Conductivity in ionic Solids, Solid held together by covalent bonding	Chapter 4: p72-114
	<i>Learning Outcomes:</i> (i) Heart of modern inorganic chemistry: structure and bonding (ii) Methods of rationalizing and predicting structures (iii) Explain how ionic and covalent bonds are formed between atoms (iv) Name molecular and ionic compounds and compare/contrast the properties of molecular and ionic compounds		
2 (5-7)	Chemical Bonding: Covalent Bond	Valence Bond Theory; Molecular Orbital Theory	Chapter 5: p117-155
	<i>Learning Outcomes:</i> (i) Preliminary analysis of covalent bonding (ii) A simple rule for predicting molecular structure (iii) A brief understanding on symmetry		
3 (8-12)	Concepts of inorganic chemistry	Electronegativity, Acid-base chemistry, A generalized acid-base concept, Measures of acid-base strength	Chapter 5: p 155-169 Chapter 8: p 220-243
	<i>Learning Outcomes:</i> (i) To define electronegativity, advances in electronegativity (ii) Concept of acid and bases, importance of different forces between atoms and ions in determining chemical properties (iii) Important aspects of each type of forces: relative strength, how rapidly it decreases with increasing distance, directional or not		
4 (13-15)	Various concepts of inorganic chemistry	Chemistry of aqueous and nonaqueous solvents, Summary of protonic and aprotic solvents, Molten salts	Chapter 9: p 245-260
	<i>Learning Outcomes:</i> (i) To analyze the solvation effects and anomalies in acid-base chemistry (ii) Identification of conjugate acid-base pairs		





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5 (16-19)	Concepts of redox chemistry	Electrode potentials, electromotive forces, Trends in standard potentials, Nernst Equation, Redox Stability, Latimer and Frost diagram, Ellingham diagram, Electrochemistry of the halogens and pseudohalogens	Chapter 9: p 260-265 Chapter 12: p. 367; Chapter 13: p. 379-389
	<i>Learning Outcomes:</i> (i) Clear and complete idea of the electronic structure of elements and chemical reactions (ii) A firm and quantitative basis of oxidation and reduction processes through electrochemical reactions (iii) To discuss complete idea of Latimer, Frost, and Ellingham diagram, different oxidation state and EMFs of different groups		
6 (20-22)	The chemistry of the main group elements: periodicity	Periodicity: First and second row anomalies, The use of <i>p</i> orbitals in pi-bonding, The use of <i>d</i> orbitals by nonmetals, Periodic anomalies of the nonmetals and post-transition metals	Chapter 10: p267-290
	<i>Learning Outcomes:</i> (i) Fascinating aspect of inorganic chemistry: diversity of reactions and structures encountered in the chemistry (ii) To systematize elemental relationships in the periodic table (iii) Describe the arrangement of elements in the periodic table and relate the arrangement to electronic configuration, bonding, and properties.		
7 (23-32)	Inorganic chains, rings, and cages	Catenation, Heterocatenation, Silicate minerals, Intercalation chemistry, One dimensional conductors, Isopoly anions, Heteropoly anions; Borazines, Phosphazenes, Phosphazene polymers, Other heterocyclic inorganic systems, Homocyclic inorganic systems; Boron cage compounds- Boranes, Carboranes, Metallacarboranes, Structure prediction for heteroboranes and organometallic clusters	Chapter 11: p292-338
	<i>Learning Outcomes:</i> (i) Chemistry of nonmetals: their propensity to form chains, rings, and cages (ii) Most metals show less tendency to form compounds of this type, and the length of the chains and size of the ring thus formed are restricted (iii) A complete and detailed discussion on different metals which show tendency to form compounds of this type		
8 (32-36)	The chemistry of halogens and the noble gases	Noble gas chemistry: Fluorides, bonding other compounds of xenon, Bond strengths in noble gas compounds, Halogens in positive oxidation states, Polyhalide ions, Fluorine-oxygen chemistry, Oxyacids of heavier halogens, Halogen oxides and oxyfluorides, Halogen cations, Halides, Pseudohalogens	Chapter 12: p343-367
	<i>Learning Outcomes:</i> (i) Discuss physical and chemical properties of halogens (ii) Different limiting factor with regard to which noble gas compounds form (iii) To discuss high electronegativity, activity of the halogens, high effective nuclear charge (iv) Harmful effect of halogens to biological organisms in sufficient quantities		





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9 (38-40)	Inorganic clusters	Metal clusters, Dinuclear compounds, Trinuclear clusters, Tetranuclear clusters, Hexanuclear clusters, Polyatomic Zintl anions and cations, Chevrel phases, Infinite metal chains	Chapter 13: p395-406
<i>Learning Outcomes:</i> (i) Properties of individual metals in various oxidation states (ii) Stability of different oxidation states, similarity, and differences (iii) Complete discussion on metal clusters, binuclear, trinuclear, tetranuclear, and hexanuclear clusters			

6. Evaluation scheme:

Component	Duration	Weightage (%) (Marks)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 min.	30 % (60)	07/10 FN1	Close Book
Comprehensive Examination	3 h	40 % (80)	08/12 FN	Close Book and Open Book
Continuous Evaluation*	25 min each	20 % (40)	Continuous	Close Book
Continuous Evaluation**	15 min each	10 % (20)	Continuous	Close Book

* In continuous evaluation, THREE tutorial tests (20 Marks each) will be conducted in entire semester. Best TWO will be considered for final evaluation.

**In continuous evaluation linked to attendance, THREE tutorial tests (10 Marks each) will be conducted in the entire semester. The tutorial tests will be graded only if the student maintains a minimum of 50% attendance. Best TWO will be considered for final evaluation.

7. Minimum criterion for awarding valid grades: Where a student continued to remain registered in a course but gave the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank script (answer book) this event will be reported as NC (Not Cleared). **Moreover, a student should obtain 30% of the average of the top 10 performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course.** If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.

8. Chamber Consultation Hour: Monday, 6-7 pm.

9. Notices: Notices, if any, will be displayed on Nalanda website.

10. Make-up Policy: There will not be any makeup for any continuous evaluations. Make up for mid sem and compre will be allowed for genuine cases as per institute guidelines.

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